

Homework 4

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Problem 1

According to example 3 we make assumptions:

$$\varepsilon_{eff} \approx \varepsilon_r$$

$$\Delta L \approx 0$$

then we can use the approximation:

$$\frac{L}{\lambda_0} \approx \frac{1}{2\sqrt{\varepsilon_r}}$$

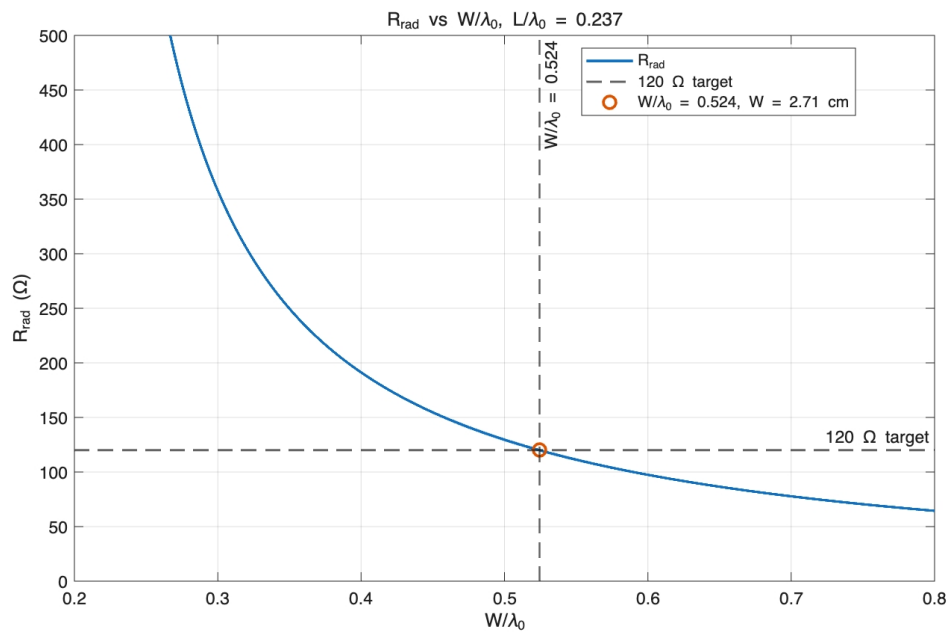
with

$$\lambda_0 = \frac{3 \times 10^8}{5.8 \times 10^9} = 5.17 \text{ cm}$$

and we get

$$\frac{L}{\lambda_0} \approx \frac{1}{2\sqrt{4.45}} = 0.237$$

We use Grad_patch_polyapp to sweep $\frac{L}{\lambda_0}$:



we get

$$W = 2.71 \text{ cm}$$

So

$$\frac{W}{h} = 17.484$$

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12h/W}} = 3.954$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}} = 2.6 \text{ cm}$$

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{eff} + 0.3)(W/h + 0.264)}{(\epsilon_{eff} - 0.258)(W/h + 0.8)} = 0.462$$

$$\Delta L = 0.072 \text{ cm}$$

$$L_{eff} = \frac{\lambda_g}{2} = 1.3 \text{ cm}$$

$$L = L_{eff} - 2\Delta L = 1.156 \text{ cm}$$

$$\frac{L}{\lambda_0} = 0.224$$

We then use Grad_patch_polyapp with

$$\frac{W}{\lambda_0} = 0.524$$

$$\frac{L}{\lambda_0} = 0.224$$

Finally we get

$$R = 116.93 \Omega$$

which is an acceptable value.

In conclusion, the length and width of the patch antenna are:

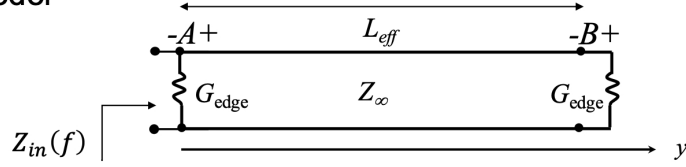
$$L = 1.156 \text{ cm}$$

$$W = 2.71 \text{ cm}$$

Problem 2

1)

TL model



We have

$$G_{edge} = \frac{1}{2} G_{rad} = \frac{1}{2R_{rad}}$$

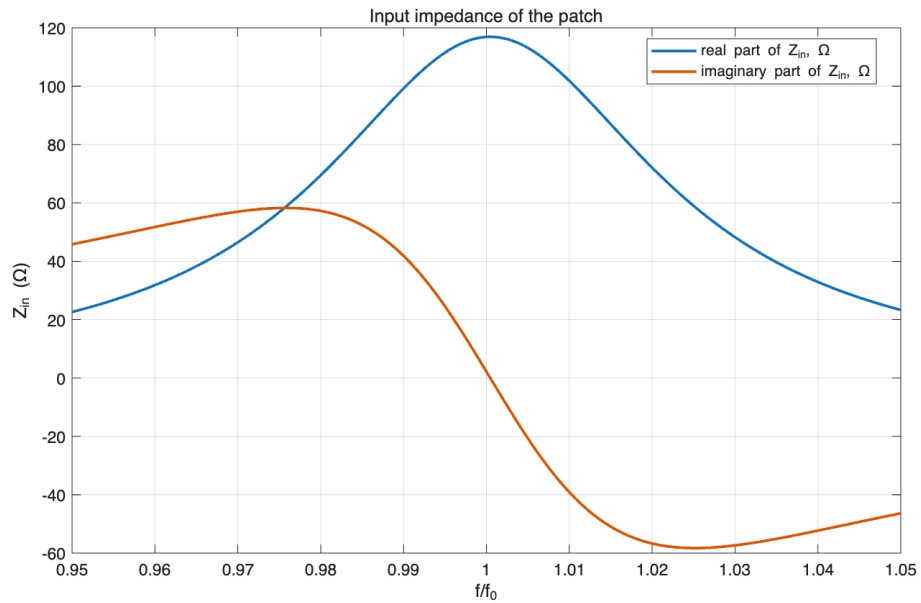
$$\Gamma_B = \frac{1/G_{edge} - Z_\infty}{1/G_{edge} + Z_\infty}$$

$$\Gamma_{A+}(f) = \Gamma_B e^{-2jk(f)L_{eff}}$$

$$Z_{in}(f) = Z_\infty \frac{1 + \Gamma_{A+}(f)}{1 - \Gamma_{A+}(f)} \parallel \frac{1}{G_{edge}}$$

$$Z_\infty = \frac{120\pi}{\sqrt{3.954(17.484 + 1.393 + 0.667 \ln(17.484 + 1.444))}} = 9.098 \Omega$$

we use Matlab to plot the real and imaginary part of $Z_{in}(f)$ using the values we calculated in **Problem 1**:



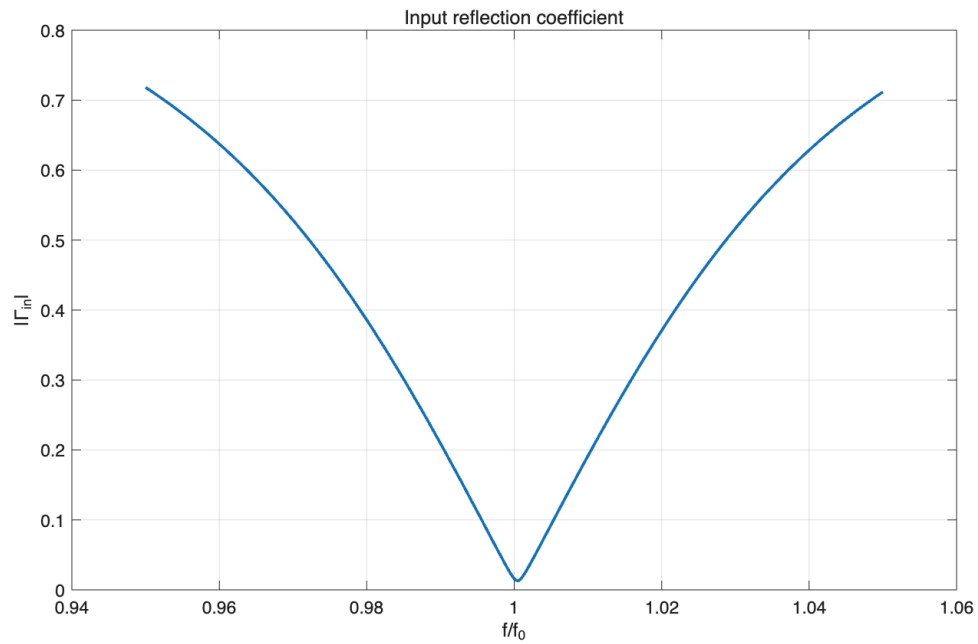
2)

We have

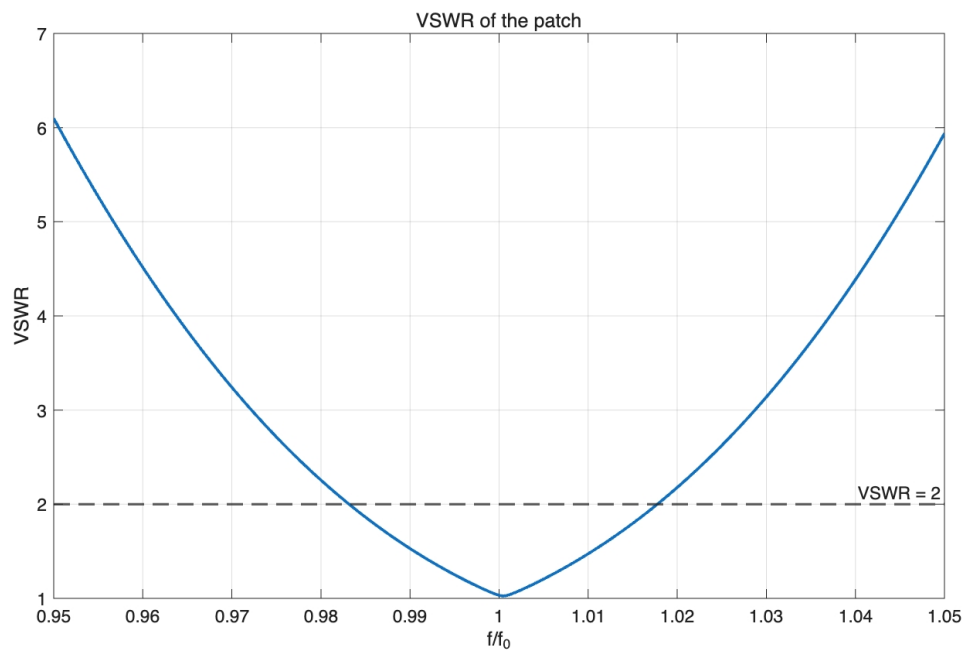
$$\Gamma_{in}(f) = \frac{Z_{in}(f) - Z_0}{Z_{in}(f) + Z_0}$$

$$Z_0 = R_0 = 120 \, \Omega$$

We use Matlab to plot $|\Gamma_{in}(f)|$ and VSWR:



At resonance $f = f_0$, $|\Gamma_{in}(f)|$ reaches its minimum value, indicating good impedance matching.



We read from Matlab results:

$$f_{low} = 5.703 \text{ GHz}, \quad f_{high} = 5.903 \text{ GHz}$$

$$BW = 200.10 \text{ MHz}$$

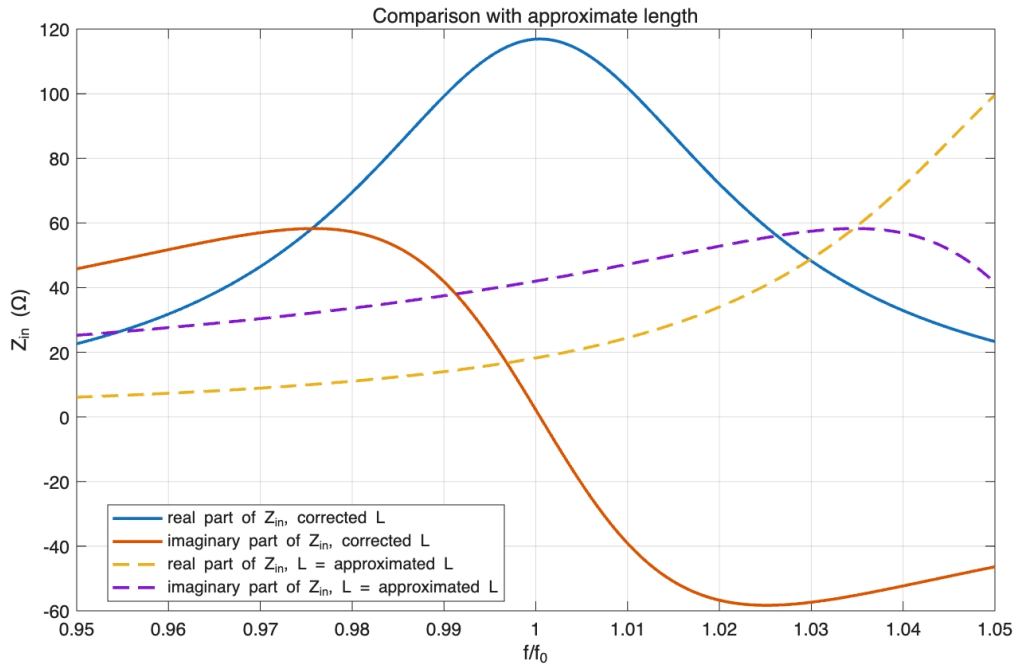
$$BW_{\%} = 3.45\%$$

3)

From **Problem 1** we get

$$L = 0.237\lambda_0 = 1.225 \text{ cm}$$

then we repeat 1) in Matlab and plot the figure:



The use of the approximate length $L = 1.225 \text{ cm}$, without considering the excess length ΔL , leads to a shift of the resonance frequency.

In particular, the condition $\text{Im} \{Z_{in}\} = 0$ is no longer satisfied at $f=f_0$, indicating that the antenna is no longer matched at the desired frequency.

This shows that the fringing field effect, represented by ΔL , has a significant impact on the accurate design of the patch antenna.

Problem 3

Using the approximated model

$$\frac{y_{TL}}{L_{eff}} = \frac{1}{\pi} \cos^{-1} \sqrt{\frac{R_{in}}{R_{rad}(0)}} = \frac{1}{\pi} \cos^{-1} \sqrt{\frac{50}{116.930}} = 0.273$$

$$y_{TL} = 0.273 \times 1.3 = 0.355 \text{ cm}$$

$$y = y_{TL} - \Delta L = 0.355 - 0.072 = 0.283 \text{ cm}$$